

Lecture: Data Independence in DBMS

Physical and logical independence in database systems

Prepared for a DBMS lecture

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Learning Objectives

After completing this lecture, students should be able to:

- 1 Explain the concept of **data independence** in DBMS.
- 2 Describe the three-level DBMS architecture: **Internal Level**, **Conceptual Level**, and **View Level**.
- 3 Distinguish between **Physical Data Independence** and **Logical Data Independence**.
- 4 Analyze practical examples of each type of data independence.
- 5 Explain why data independence reduces maintenance effort and supports long-term system evolution.

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What Is Data Independence?

Definition

Data independence is the ability to change the schema at one level of a database architecture without affecting the schema at the next higher level.

In simpler terms:

- The internal storage or organization of data may change.
- Users and application programs can still access data as before.
- Application code does not need major changes after internal database optimizations.

Introductory Example

Suppose a sales system frequently searches orders by customer ID.

The database administrator creates an index to speed up the query:

- The internal access path changes.
- Users still use the same application interface.
- The application still uses the same SQL query.

Key Idea

The DBMS hides internal database changes from users and application programs.

Why Is Data Independence Important?

Lower Maintenance

Developers do not need to update applications after every small database change.

Greater Flexibility

The database can be reorganized or optimized internally with limited impact on users.

Long-Term Evolution

The database can be extended as business requirements change without breaking existing systems.

Quick Quiz: Concept of Data Independence

Question 1. What is data independence in DBMS?

- ☐ A. The ability to change a schema at one level without affecting the next higher level
- ☐ B. The ability to delete all data from the system
- ☐ C. The ability to change the user interface color
- ☐ D. The ability to replace SQL with HTML

Quick Quiz: Role of Data Independence

Question 2. Why does data independence reduce maintenance effort?

- ☐ A. Developers do not need to modify applications after every small database change
- ☐ B. Users must manage disk storage themselves
- ☐ C. The database can never be changed
- ☐ D. A DBMS no longer needs a schema

Question 3. How does data independence support long-term development?

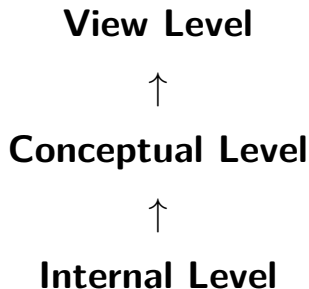
- ☐ A. It allows the database to be updated and extended with less risk of breaking existing systems
- ☐ B. It requires the whole application to be rewritten every year
- ☐ C. It prevents all changes to data structure
- ☐ D. It works only with text files

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Three-Level DBMS Architecture

To understand data independence, we need to understand the three levels of DBMS architecture:



Main Idea

Each level hides details of the lower level to reduce dependency between applications and physical data storage.

Definition

The **Internal Level** describes how data is physically stored in the system.

It includes:

- Storage files.
- Indexes.
- Disk space allocation.
- Data compression.
- Data access methods.
- Internal storage structures.

Conceptual Level

Definition

The **Conceptual Level** describes the logical structure of the database.

It includes:

- Tables and fields.
- Attributes.
- Relationships among tables.
- Primary keys and foreign keys.
- Integrity constraints.

Definition

The **View Level** describes how users and applications see the data.

Examples:

- Students can view only their own grades.
- Lecturers can view students in their assigned classes.
- The academic office can view academic records.
- The finance office can view tuition-related data.

Three Levels and Data Independence

Level	Main Concern	Typical Users
View	User-specific data views	End users, applications
Conceptual	Logical database design	Designers, developers
Internal	Physical storage details	DBA, DBMS engine

Important

Data independence is achieved by separating these levels and maintaining mappings among them.

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Two Types of Data Independence

Type	Changes Occur At	Should Not Affect
Physical Data Independence	Internal Level	Conceptual Level and application programs
Logical Data Independence	Conceptual Level	View Level and application programs

Observation

Physical data independence is usually easier to achieve than logical data independence.

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Physical Data Independence: Definition

Definition

Physical Data Independence is the ability to change how data is physically stored without affecting the logical schema or user applications.

It is related to changes at the **Internal Level**.

Main Focus

Physical data independence focuses on storage, performance, indexing, and access paths.

Examples of Physical Changes

The following changes are examples of physical data independence:

- Moving database files from one disk to another.
- Migrating from HDD to SSD.
- Creating indexes to speed up queries.
- Compressing data files to save storage space.
- Changing access paths or data access methods.
- Changing the internal file organization.

SQL Example: Creating an Index

Assume the system frequently searches orders by `customer_id`. The DBA creates an index:

```
CREATE INDEX idx_orders_customer  
ON Orders(customer_id);
```

The application can still use the same query:

```
SELECT *  
FROM Orders  
WHERE customer_id = 101;
```

The logical schema of `Orders` does not change.

Benefits of Physical Data Independence

- Allows backend optimization without changing user applications.
- Improves query performance through indexes and better access paths.
- Supports storage and hardware upgrades.
- Reduces the need to modify application programs.
- Improves long-term maintainability.

Quick Quiz: Physical Data Independence

Question 1. What is Physical Data Independence?

- ☐ A. The ability to change physical storage without affecting the logical schema or application
- ☐ B. The ability to change the user interface
- ☐ C. The ability to delete the whole database
- ☐ D. The ability to rename a course

Question 2. Which example illustrates Physical Data Independence?

- ☐ A. Creating an index to improve query speed without changing the application
- ☐ B. Adding an email field to a user form
- ☐ C. Changing the website color
- ☐ D. Removing a button from the interface

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Logical Data Independence: Definition

Definition

Logical Data Independence is the ability to change the logical schema, that is, the structure and relationships of data, without affecting views or application programs.

It is related to changes at the **Conceptual Level**.

Main Focus

Logical data independence focuses on tables, attributes, relationships, and views.

Examples of Logical Changes

The following changes are examples of logical data independence:

- Adding a new column to a table.
- Removing an unused column.
- Creating a new relationship between tables.
- Splitting one table into several tables.
- Merging multiple tables.
- Creating a view to simplify data access.

SQL Example: Adding a Column

Suppose the Employees table initially contains:

```
employee_id, full_name, department
```

The company wants to store employee email addresses:

```
ALTER TABLE Employees  
ADD email VARCHAR(100);
```

If old applications use only `employee_id`, `full_name`, and `department`, they can continue to work.

Role of Views

When the logical schema changes, a view can help preserve a stable data interface for applications.

```
CREATE VIEW EmployeeBasicView AS  
SELECT employee_id, full_name, department  
FROM Employees;
```

Old applications can query EmployeeBasicView without knowing that Employees now contains an extra column.

Benefits of Logical Data Independence

- Makes application code easier to maintain.
- Supports system updates as business requirements evolve.
- Reduces the need to rewrite old queries.
- Allows the database schema to evolve more flexibly.
- Supports multiple views for different user groups.

Quick Quiz: Logical Data Independence

Question 1. What is Logical Data Independence?

- ☐ A. The ability to change the logical schema without affecting views or application programs
- ☐ B. The ability to change the physical file location
- ☐ C. The ability to move data from HDD to SSD
- ☐ D. The ability to change the operating system

Question 2. Which example illustrates Logical Data Independence?

- ☐ A. Adding an email column to Employees while old applications still work
- ☐ B. Creating an index on disk
- ☐ C. Moving data to an SSD
- ☐ D. Compressing a physical data file

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Physical vs. Logical Data Independence

Criterion	Physical Data Independence	Logical Data Independence
Main focus	Physical data storage	Logical data structure
Related level	Internal Schema	Conceptual Schema
Typical changes	Files, indexes, disks, compression, access paths	Tables, columns, relationships, views
Impact on applications	Should not affect application programs	Harder to avoid application impact
Main goal	Improve performance and storage	Evolve database design
Difficulty	Usually easier	Usually harder
Example	Moving files or adding an index	Adding or removing a column

Integrated Example: Sales System

Consider a sales management system.

Physical Data Independence: The DBA creates an index on `customer_id` in `Orders`.

```
CREATE INDEX idx_customer_id  
ON Orders(customer_id);
```

The old query still works:

```
SELECT *  
FROM Orders  
WHERE customer_id = 1001;
```

Integrated Example: Logical Change

Logical Data Independence: The company wants to store customer email addresses.

```
ALTER TABLE Customers  
ADD email VARCHAR(100);
```

Old reports that use only `customer_id` and `customer_name` can still work.

A view can preserve the old structure:

```
CREATE VIEW CustomerBasicView AS  
SELECT customer_id, customer_name  
FROM Customers;
```

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Review Questions: Multiple Choice I

Question 1. What is data independence?

- ☐ A. The ability to change a schema at one level without affecting the next higher level
- ☐ B. The ability to delete data without control
- ☐ C. The ability to change the interface color
- ☐ D. The ability to replace a database with an image file

Question 2. How many main types of data independence are there?

- ☐ A. 1
- ☐ B. 2
- ☐ C. 3
- ☐ D. 4

Review Questions: Multiple Choice II

Question 3. Physical Data Independence is mainly related to which level?

- ☐ A. Internal Level
- ☐ B. View Level
- ☐ C. User Interface
- ☐ D. Application Screen

Question 4. Logical Data Independence is mainly related to which level?

- ☐ A. Internal Level
- ☐ B. Conceptual Level
- ☐ C. Disk Level
- ☐ D. Hardware Level

Review Questions: Multiple Choice III

Question 5. Which example illustrates Physical Data Independence?

- ☐ A. Creating a new index without changing the application
- ☐ B. Adding an email field to the user interface
- ☐ C. Changing the website color
- ☐ D. Removing a button from the screen

Question 6. Which example illustrates Logical Data Independence?

- ☐ A. Adding a new column to a table while old applications still work
- ☐ B. Moving data to an SSD
- ☐ C. Creating a physical file
- ☐ D. Changing the disk block size

Short-Answer Questions

- 1 Explain the concept of data independence in DBMS.
- 2 Distinguish between Physical Data Independence and Logical Data Independence.
- 3 Why can Physical Data Independence improve performance without affecting applications?
- 4 Why is Logical Data Independence usually harder to achieve than Physical Data Independence?
- 5 Explain the role of views in supporting Logical Data Independence.

Applied Exercises

- 1 A DBA moves data from HDD to SSD and creates an index to speed up queries. The application does not change. Identify the type of data independence and explain.
- 2 An HR system adds an `email` column to `Employees`. Old reports still show only employee ID, name, and department. Identify the type of data independence and explain.
- 3 A sales system changes the structure of `Customers`, but old applications still need `customer_id` and `customer_name`. Propose a view-based solution.
- 4 Give one student management system example for each type of data independence.

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Lesson Summary

- Data independence allows schema changes at one level without affecting the next higher level.
- DBMS architecture usually has three levels: Internal, Conceptual, and View.
- Physical Data Independence allows changes in physical storage without affecting the logical schema or applications.
- Logical Data Independence allows changes in the logical schema without affecting views or applications.
- Physical Data Independence is usually easier to achieve than Logical Data Independence.
- Views can help hide logical schema changes and keep applications stable.

Key Terms

- Data Independence
- Physical Data Independence
- Logical Data Independence
- DBMS
- Internal Level
- Conceptual Level
- View Level
- Internal Schema
- Conceptual Schema
- View
- Index
- Schema
- Database Administrator
- Application Program

Closing Thought

Core Message

Data independence is one of the key reasons why DBMSs are more powerful and maintainable than direct file-based data management.

A well-designed database system separates:

- how data is physically stored,
- how data is logically organized,
- and how users or applications see the data.